AWE NUCLEAR SECURITY TECHNOLOGIES

EPOCH Workshop 2026: Fields, lasers and boundaries

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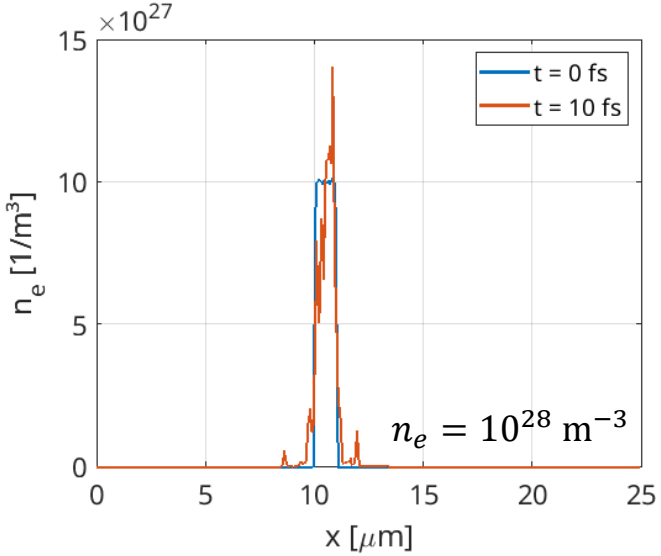
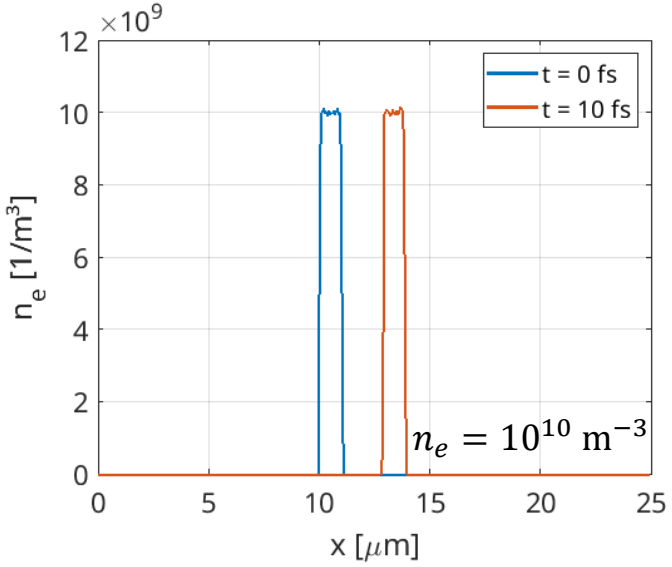
Weird result from Example 1.1

- Electron bunch motion varies density
  - 1 MeV bunch, $10^{28}\ \mathrm{m}^{-3}$ doesn't move
  - 1 MeV bunch, $10^{10}\ \mathrm{m}^{-3}$ does move

Why won't our dense electron bunch move?





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Discretisation

- Maxwell's equations:

$$\frac{\partial \boldsymbol{E}}{\partial t} = c^2 \nabla \times \boldsymbol{B} - \frac{1}{\epsilon_0}\boldsymbol{J}$$
$$\frac{\partial \boldsymbol{B}}{\partial t} = -\nabla \times \boldsymbol{E}$$

Time evolution

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\epsilon_0}$$
$$\nabla \cdot \boldsymbol{B} = 0$$

Initial conditions

First order method:

$$\frac{\partial f}{\partial t} \approx \frac{\Delta f}{\Delta t}$$

$$\Delta \mathbf{E} = \left(c^2 \nabla \times \boldsymbol{B} - \frac{1}{\epsilon_0}\boldsymbol{J} \right) \Delta t$$
$$\Delta \boldsymbol{B} = -(\nabla \times \boldsymbol{E}) \Delta \mathrm{t}$$

- To proceed, we must evaluate spatial and temporal differences in fields
  - What does this actually mean?

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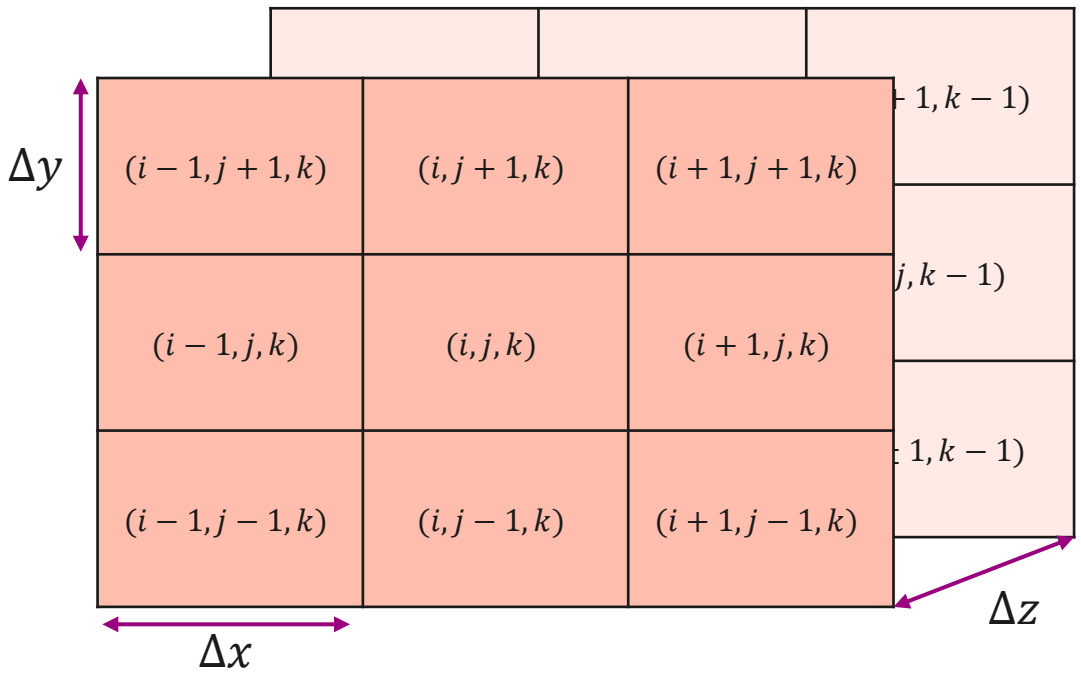
Spatial mesh

- Need to solve terms like:
  - $(\nabla \times f)_x \approx \frac{\Delta f}{\Delta y} - \frac{\Delta f}{\Delta z}$
- Find Δf:
  - Split up simulation space into cells
  - Assign a field value to each cell
  - Subtract neighbouring field values to get Δf
- Position of evaluation points for f is important!



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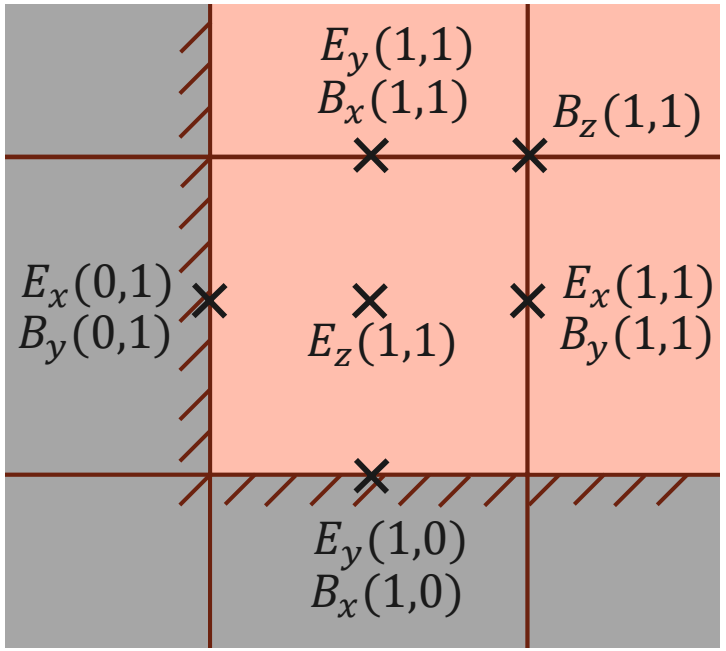
Spatial mesh

- For second-order accuracy, we can use Yee-stagger:
  - Evaluate fields at different points in each cell

- Off-set rules:
  - E: shift up a half-cell in the component direction
  - B: shift up a half-cell in every direction except component direction
  - 1D/2D: ignore shift in omitted dimensions
  - J: Evaluate current at the same point as E



- Gradients are evaluated at the mid-point between field points
  - E.g. the E_z update needs $\frac{\partial B_y}{\partial x}$ and $\frac{\partial B_x}{\partial y}$
  - Field points are staggered from E_z, these gradients are on E_z

$$\left(\frac{\Delta B_y}{\Delta x}\right)_{i,j,k} = \frac{B_y(i,j,k) - B_y(i-1,j,k)}{\Delta x}$$

$$\left(\frac{\Delta B_x}{\Delta y}\right)_{i,j,k} = \frac{B_x(i,j,k) - B_x(i,j-1,k)}{\Delta y}$$

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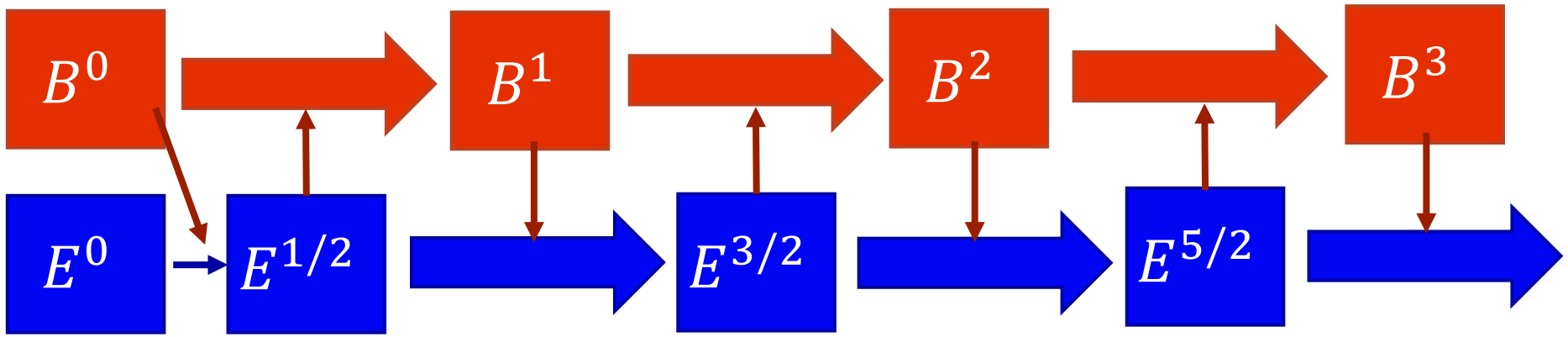
Leap-frog solver

- Second-order solver, use time-centred fields:

$$E_x^{n+1/2} = E_x^{n-1/2} + \Delta t \left(c^2 \left(\frac{\Delta B_z^n}{\Delta y} - \frac{\Delta B_y^n}{\Delta z} \right) - \frac{1}{\epsilon_0} J_x^n \right) \qquad B_x^{n+1} = B_x^n - \Delta t \left(\frac{\Delta E_z^{n+1/2}}{\Delta y} - \frac{\Delta E_y^{n+1/2}}{\Delta z} \right)$$

- Time-step Δt restricted by CFL condition: $c\Delta t < \min(\Delta x, \Delta y, \Delta z)$
- Create half-time-step displacement using first-order update for the first step



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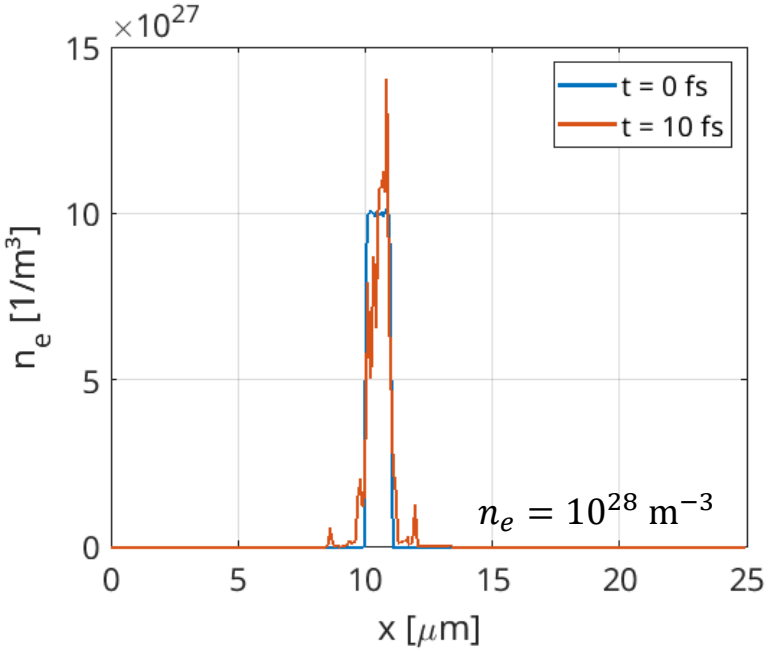
Example 1.1 explained

- The EPOCH field-solver initialises:
  - $\boldsymbol{E} = 0$
  - $\boldsymbol{B} = 0$
- Field solver is only sensitive to current $\boldsymbol{J}$
  - Absolute charge is ignored!
- $E = 0$ implies initially all cells are neutral
  - The $10^{28}\ \mathrm{m}^{-3}$ bunch ran as if a dense, neutralising ion bunch was also present
  - This “fake bunch” trapped our 1 MeV e^-



$$E_x^{n+1/2} = E_x^{n-1/2} + \Delta t \left(c^2 \left(\frac{\Delta B_z^n}{\Delta y} - \frac{\Delta B_y^n}{\Delta z} \right) - \frac{1}{\epsilon_0} J_x^n \right)$$

$$B_x^{n+1} = B_x^n - \Delta t \left(\frac{\Delta E_z^{n+1/2}}{\Delta y} - \frac{\Delta E_y^{n+1/2}}{\Delta z} \right)$$

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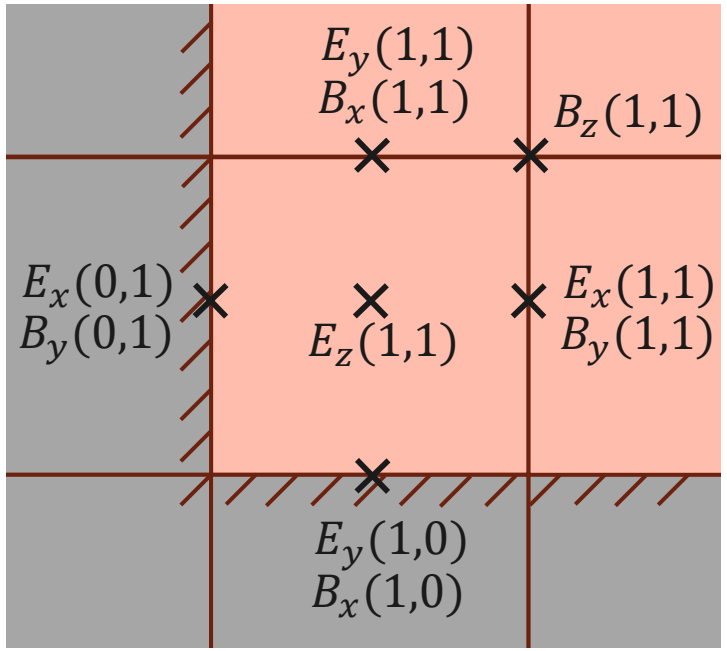
Boundary conditions

- EPOCH simulation domain is surrounded by ghost-cells
  - Used to average E, B fields on near-edge particles

- Two basic field boundaries:



Reflect

- Staggered fields:
  - $f(0) = 0$
  - $f(-1) = f(1)$
  - $f(-2) = f(2)$

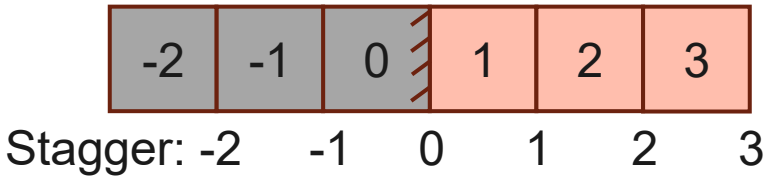
- Cell-centre fields:
  - $f(0) = f(1)$
  - $f(-1) = f(2)$
  - $f(-2) = f(3)$

Periodic

- Let n_x be the last cell calculated normally

- All fields:
  - $f(0) = f(n_x)$
  - $f(-1) = f(n_x - 1)$
  - $f(-2) = f(n_x - 2)$

  - $f(n_x + 1) = f(1)$



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Example 2.1: Two stream instability

- Two plasma beams pass through each other in a periodic domain
- Noise in currents generates fields in field solver
- Phase space of particles start to “swirl”
- EPOCH can directly output the phase-space of plasma particles

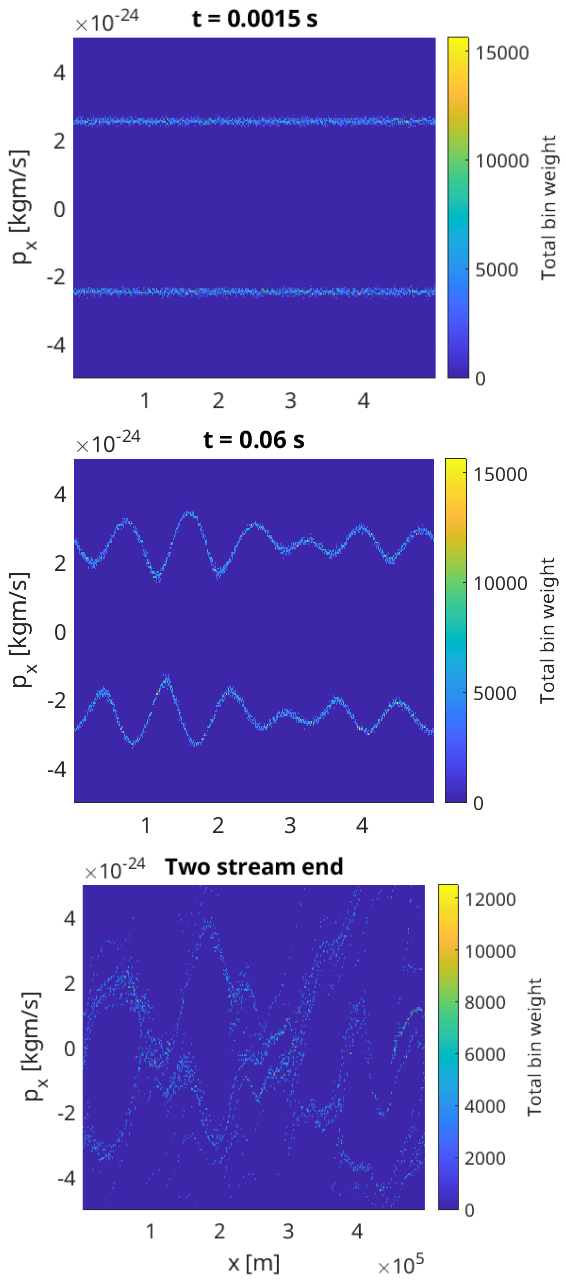
```
begin:dist_fn
  name = x_px
  ndims = 2

  direction1 = dir_x
  direction2 = dir_px

  # Range is ignored for spatial coordinates
  range1 = (1, 1)
  range2 = (-5e-24, 5e-24)

  # Resolution is ignored for spatial coordinates
  resolution1 = 1
  resolution2 = 200

  include_species:Left
  include_species:Right
end:dist_fn
```



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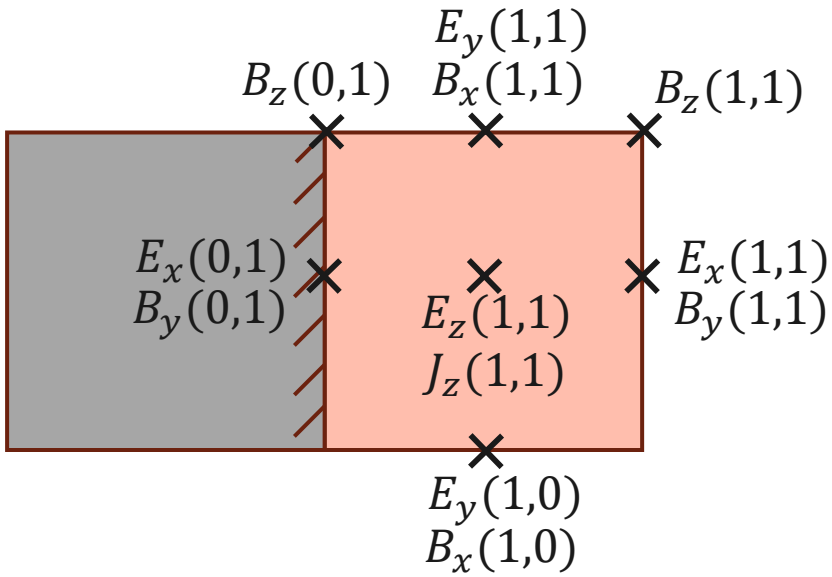
Laser boundary



- Special boundaries are used for lasers
  - Can do two-sided gradient for $E_x(0,1)$, but $B_y(0,1)$ and $B_z(0,1)$ need special treatment
  - Consider $B_y(0,1)$ condition, in a convoluted way...

- Firstly, consider the time-gradient:
  - $\left(\frac{\partial E_z}{\partial t}\right)^n \approx \frac{1}{\Delta t}\left(E_z^{n+1/2} - E_z^{n-1/2}\right)$
- Time averaging condition:
  - $\boldsymbol{E}^n \approx \frac{1}{2}\left(\boldsymbol{E}^{n+1/2} + \boldsymbol{E}^{n-1/2}\right)$
  - $\boldsymbol{E}^{n+1/2} = 2\boldsymbol{E}^n - \boldsymbol{E}^{n-1/2}$
- Hence:
  - $\left(\frac{\partial E_z}{\partial t}\right)^n \approx \frac{1}{\Delta t}\left(2E_z^n - 2E_z^{n-1/2}\right)$

- Field solver doesn't have E_z^n though!
  - But we do have $\boldsymbol{B}^n$ …
- Assume inward plane waves, so $E_z = -cB_y$
- Let $S = E_z = \frac{1}{2}\left(E_z - cB_y\right)$, so:
  - $\left(\frac{\partial E_z}{\partial t}\right)^n = \frac{1}{\Delta t}\left(4S^n + 2B_y^n - 2E_z^{n-1/2}\right)$

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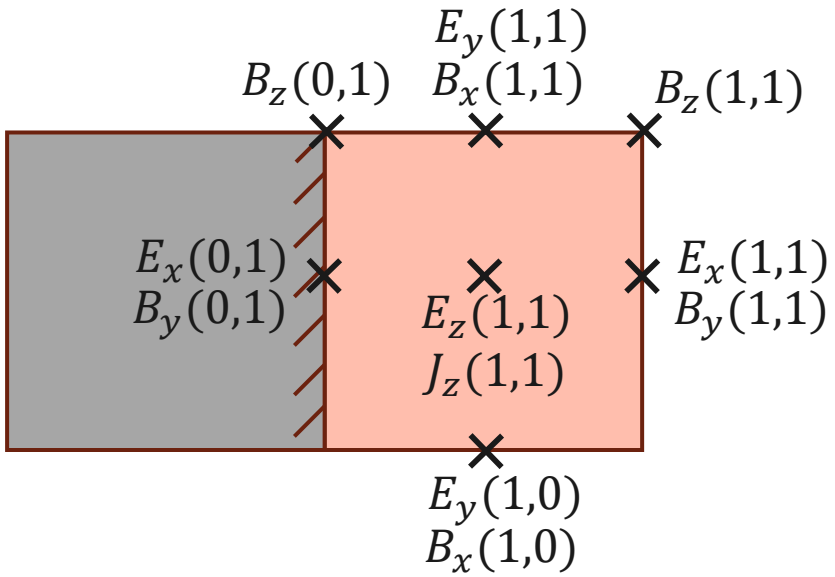
Laser boundary continued



- So we have:
  - $\left(\frac{\partial E_z}{\partial t}\right)^n = \frac{1}{\Delta t}\left(4S^n + 2B_y^n - 2E_z^{n-1/2}\right)$
- Ampere-Maxwell:
  - $\left(\frac{\partial E_z}{\partial t}\right)^n = c^2 \frac{(B_y)_{1,1}^n - (B_y)_{0,1}^n}{\Delta x} - \frac{1}{\epsilon_0}(J_z)_{1,1}^n$
- Combining the two yields:
  - $(B_y)_{0,1}^n = \left(\frac{c^2}{\Delta x} + \frac{1}{\Delta t}\right)^{-1}\left(-\frac{4}{\Delta t}S^n + \left(\frac{c^2}{\Delta x} - \frac{1}{\Delta t}\right)(B_y)_{1,1}^n + \frac{2}{\Delta t}(E_z)_{1,1}^{n-\frac{1}{2}} - \frac{1}{\epsilon_0}(J_z)_{1,1}^n\right)$

Note: $\frac{\partial E_z}{\partial t}$ is evaluated at the $J_z(1,1)$ position.
There's no B_y here, so average:

$$2B_y^n \approx (B_y)_{1,1}^n + (B_y)_{0,1}^n$$

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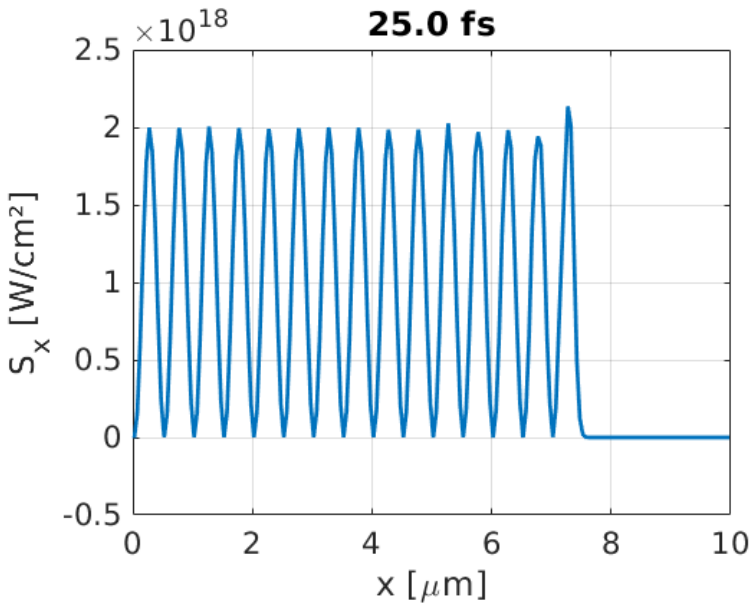
Example 2.2: Plane wave in 1D

- Inject a simple plane wave into a 1D EPOCH simulation
  - Plot the Poynting Flux:

$$\boldsymbol{S} = \frac{1}{\mu_0}(\boldsymbol{E} \times \boldsymbol{B})$$

- Front bulge:
  - Incomplete oscillation on front

- Twice the injected wavelength?
  - Peak and trough of E_y are both peaks




```
begin:control
    nx = 200
    t_end = 25e-15
    x_min = 0
    x_max = 10e-6
    stdout_frequency = 100
end:control

begin:boundaries
    bc_x_min = simple_laser
    bc_x_max = open
end:boundaries

begin:laser
    boundary = x_min
    intensity_w_cm2 = 1.0e18
    lambda = 1.0e-6
end:laser

begin:output
    dt_snapshot = t_end
    poynt_flux = always
end:output
```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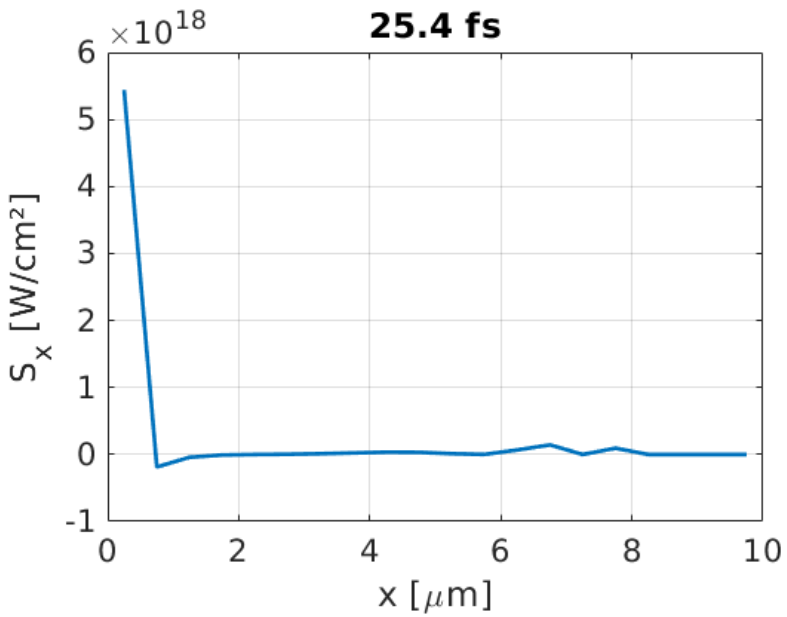
Example 2.3: Plane wave resolution

- What resolution can we get away with?
  - Last example: 200 cells
  - Now try 20 cells

- Field solver relies on gradients
- Too few evaluation points on the wave prevents this




```
begin:control
    nx = 20
    t_end = 25e-15
    x_min = 0
    x_max = 10e-6
    stdout_frequency = 100
end:control
```

Rule of thumb:
Need about 20 cells per wavelength

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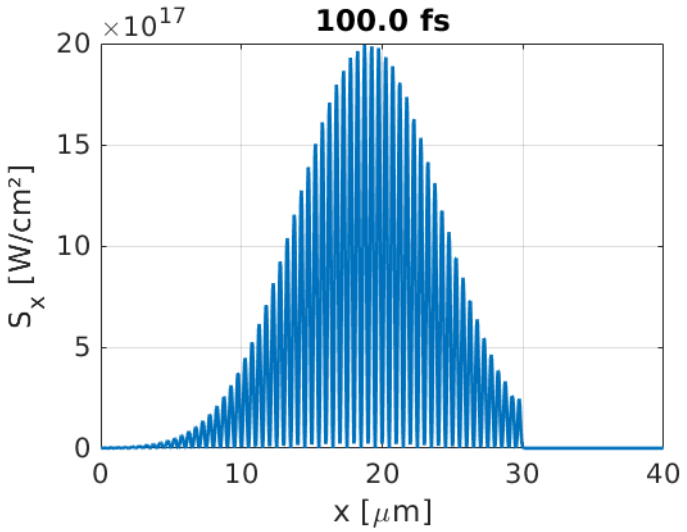
Example 2.4: Gaussian pulse in 1D

- Add a temporal profile to the laser
- EPOCH gauss function (not a true Gaussian):

$$\mathrm{gauss}(x, x_0, w) = e^{-\frac{(x-x_0)^2}{w^2}}$$

- The profile is applied to the E field amplitude
  - $E \propto I^{1/2}$
  - This function, applied to E, gives a real Gaussian in I (intensity)



Start:10% maximum intensity
Width: 50% at 13 to 25 μm
(~40 fs at c)

```
begin:control
    nx = 1000
    t_end = 100e-15
    x_min = 0
    x_max = 40e-6
    stdout_frequency = 100
end:control

begin:boundaries
    bc_x_min = simple_laser
    bc_x_max = open
end:boundaries

begin:constant
    t_fwhm = 40.0e-15
    w_t = t_fwhm / sqrt(2*loge(2))
    t_hw01m = 0.5 * t_fwhm * sqrt(loge(10)/loge(2))
end:constant

begin:laser
    boundary = x_min
    intensity_w_cm2 = 1.0e18
    lambda = 1.0e-6
    t_profile = gauss(time,t_hw01m,w_t)
end:laser

begin:output
    dt_snapshot = t_end
    poynt_flux = always
end:output
```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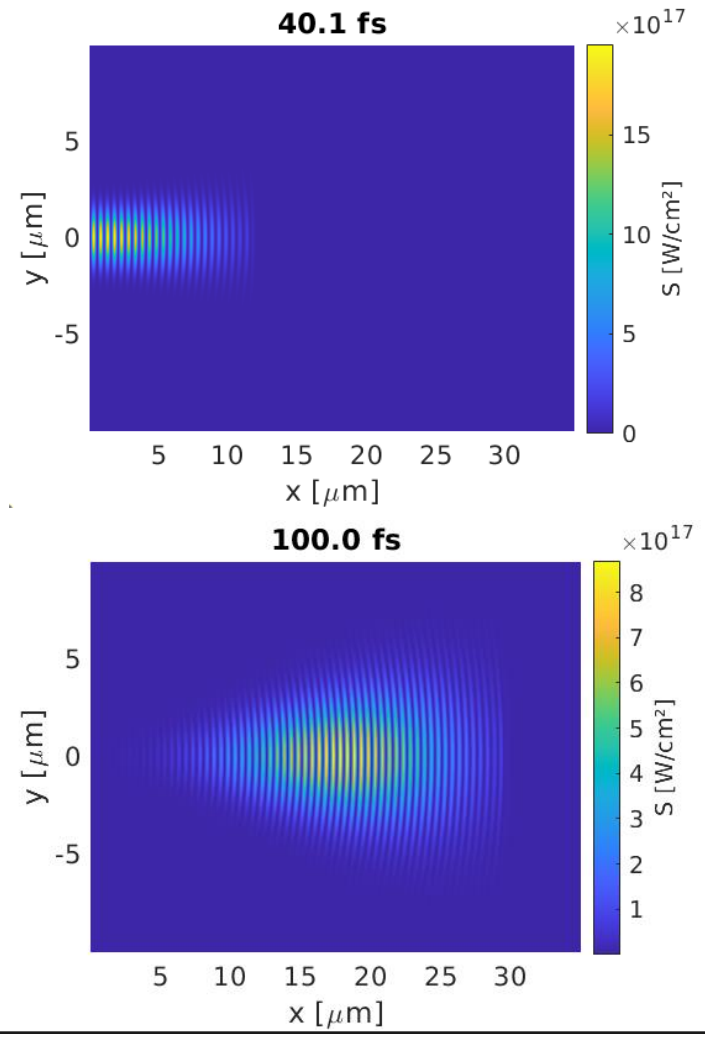
Example 2.5: Gaussian pulse in 2D

- Model a Gaussian pulse in 2D:
  - Gaussian temporal profile
  - Gaussian spatial profile

- See how it spreads out and loses intensity – why does it do this?




```
begin:control
    nx = 700
    ny = 400
    t_end = 100e-15
    x_min = 0
    x_max = 35e-6
    y_min = -10e-6
    y_max = 10e-6
    stdout_frequency = 100
end:control

begin:boundaries
    bc_x_min = simple_laser
    bc_x_max = open
    bc_y_min = open
    bc_y_max = open
end:boundaries

begin:constant
    t_fwhm = 40.0e-15
    y_fwhm = 2.0e-6
    w_t = t_fwhm / sqrt(2*loge(2))
    w_y = y_fwhm / sqrt(2*loge(2))
    t_hw01m = 0.5 * t_fwhm * sqrt(loge(10)/loge(2))
end:constant

begin:laser
    boundary = x_min
    intensity_w_cm2 = 1.0e18
    lambda = 1.0e-6
    profile = gauss(y,0,w_y)
    t_profile = gauss(time,t_hw01m,w_t)
end:laser

begin:output
    dt_snapshot = 10 * femto
    poynt_flux = always
end:output
```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Example 2.6: Gaussian beam theory

- Paraxial approximation:
  - Laser comes in and out of focus
  - Valid if minimum waist is much smaller than wavelength

- Previous example:
  - Simulated tight focus on axis
  - Divergence from this is expected

$$\boldsymbol{E}(r,x) = E_0 \frac{w_0}{w(x)} e^{-r^2/w(x)^2} e^{-i\left(kx - \psi(x) + \frac{kr^2}{2R_c(x)}\right)} \hat{\boldsymbol{y}}$$

Where:

- r is the radial distance from the laser axis
- x is axial distance from the focal position
- E_0 is the peak E field amplitude at the focus
- $w(x)$ is the beam-waist at x (radial distance where field strength drops by e^{-1})
- w_0 is $w(x = 0)$
- k is the laser wave-vector
- $R_c(x)$ is the radius of curvature at x
- $\psi(x)$ is the Gouy phase correction

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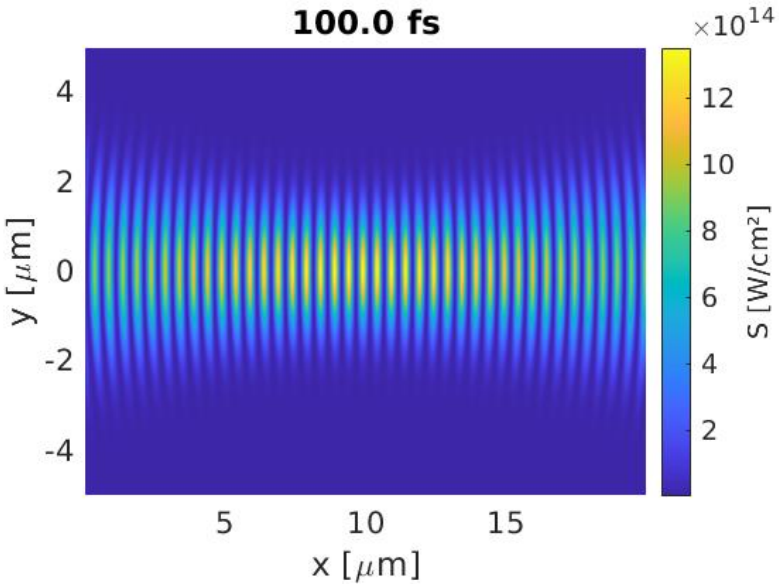
Example 2.6: Gaussian beam theory



- Setup beam on axis to focus after 10 μm

- Peak expected intensity still not reached
  - Focal spot too tight c.f. wavelength
  - Paraxial approximation at limit

$$\boldsymbol{E}(r,x) = E_0 \frac{w_0}{w(x)} e^{-r^2/w(x)^2} e^{-i\left(kx - \psi(x) + \frac{kr^2}{2R_c(x)}\right)} \hat{\boldsymbol{y}}$$

```
begin:constant
    I_fwhm = 2.0e-6          # FWHM of laser intensity
    I_peak_Wcm2 = 1.0e15     # 0.5 * eps0 * c * E_peak^2
    las_lambda = 1.0e-6      # Laser wavelength
    foc_dist = 10.0e-6        # Boundary to focal point distance
end:constant

begin:constant
    las_k = 2.0 * pi / las_lambda
    w0 = I_fwhm / sqrt(2.0 * loge(2.0))                  # Beam Waist
    ray_rang = pi * w0^2 / las_lambda                    # Rayleigh range
    w_boundary = w0 * sqrt(1.0 + (foc_dist/ray_rang)^2)  # Waist on boundary
    I_boundary = I_peak_Wcm2 * (w0 / w_boundary)^2       # Intens. on boundary
    rad_curve = foc_dist * (1.0 + (ray_rang/foc_dist)^2) # Boundary curv. rad.
    gouy = atan(-foc_dist/rad_curve)                     # Boundary Gouy shift
end:constant

begin:laser
    boundary = x_min
    intensity_w_cm2 = I_boundary
    lambda = las_lambda
    phase = las_k * y^2 / (2.0 * rad_curve) - gouy
    profile = gauss(y, 0, w_boundary)
end:laser
```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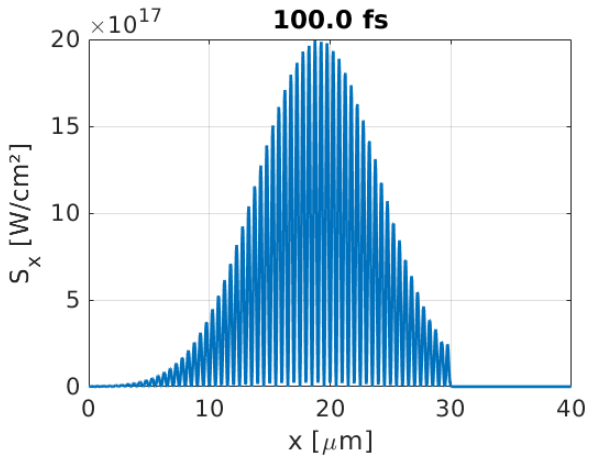
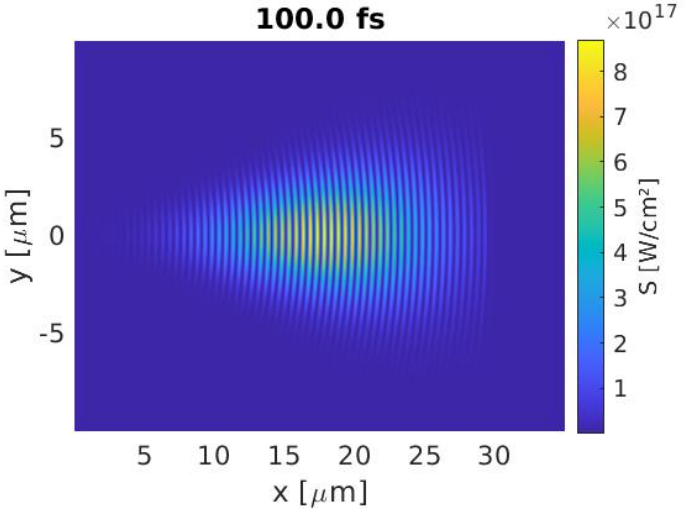
Practical sessions

- Your turn!
  - Workshop practical sessions to support for your own research
  - Demonstrators will float around to help setup/diagnostics
  - Aim to complete a small project during the workshop

- Project time:
  - 2 hours today
  - 3 hours tomorrow

- End goal:
  - Produce a slide with figures/gifs to present in a “laptop poster session”





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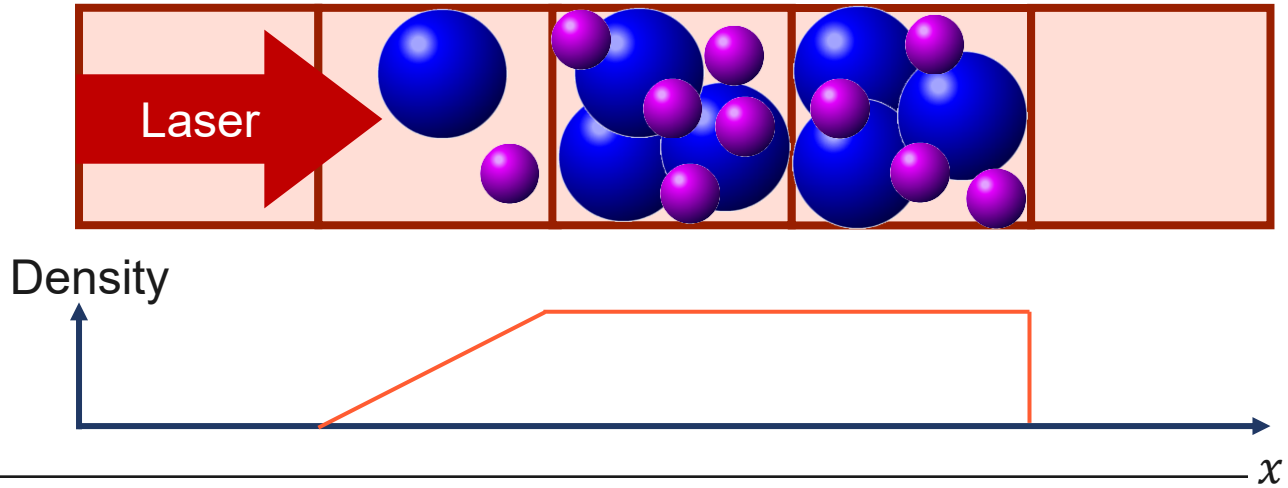
Example project

- Study energetic particles made when a high intensity laser hits a solid target, in 1D

- Laser:
  - 300 nm to 1 μm wavelength
  - 10^{20} to 10^{24} Wcm^{-2}
  - Pulse length, 30 fs to 200 fs
  - 1 laser, multiple lasers

- Target:
  - Fully/partially ionised
  - Various densities and temperatures
  - Density ramp on edges, or clean surfaces

- Output:
  - Target density/temperature evolution
  - Electron phase-space
  - Radiation energy spectra



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